WolfISMS

Wolf's Information Security Management System

Docker Installation

Version 3.63

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1. Overview

This guide describes how to run WolfISMS as a Docker container behind the Nginx Proxy Manager (NPM). NPM handles TLS termination and certificate management, while WolfISMS itself speaks plain HTTP inside the Docker network.

It is aimed at administrators who already operate a Docker environment. For the classic installation directly on an Ubuntu or Debian server with Apache and mod_wsgi, see chapter 3 of the administration documentation instead.

1.1 Target Architecture

Component	Role
Nginx Proxy Manager	Accepts HTTP (80) and HTTPS (443) from the outside, terminates TLS, manages the Let's Encrypt certificates and forwards requests to the WolfISMS container on the internal network. The administration interface runs on port 81
WolfISMS container	The application itself. It is not directly reachable from outside, only through the Docker network proxy
Database	MariaDB or MySQL – either as its own container or as an existing database server, depending on your operating model
Docker network "proxy"	Connects NPM and WolfISMS. It is created manually once and referenced as an external network by both Compose projects

1.2 Why HTTPS Is Mandatory

WolfISMS sets its session cookies with the attribute Secure=True. Browsers transmit such cookies only over encrypted connections. Over plain HTTP the login page will therefore load, but signing in fails because the session cannot be retained. A valid certificate is not a recommendation but an operating prerequisite.

Reverse proxy support is built in:
The application evaluates the headers X-Forwarded-Proto and X-Forwarded-Host (ProxyFix in isms.py). It therefore correctly recognises that it runs behind a TLS termination point and generates absolute links using https and the public host name. The Nginx Proxy Manager sets these headers by default – no additional configuration is required.

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2. Prerequisites

Component	Requirement
Docker Engine	Current version with the Compose V2 plug-in (docker compose, not docker-compose)
Operating system	Any Docker host (Linux recommended)
Public DNS record	The desired host name, e.g. isms.your-domain.com, must resolve to the public IP address of the Docker host
Open ports	Ports 80 and 443 must be reachable from the outside. Port 80 is required for the Let's Encrypt ACME challenge
Port 81	The NPM administration interface. It should not be publicly reachable but restricted to the internal network or VPN
Installation package	The WolfISMS repository or Docker package containing docker-compose.yml and .envdefault (provided by the vendor)
Database	MariaDB 10.11+ or MySQL 8.0+ – as a container or as an external server

Do not expose port 81 to the internet:
The administration interface of the Nginx Proxy Manager controls all certificates and forwarding rules. Restrict access to the internal network using the host firewall, and change the default credentials immediately after the first sign-in.

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3. Step 1 – Create the Docker Network

Both Compose projects – the Nginx Proxy Manager and WolfISMS – reference the same external network. It is created once on the Docker host:

```
docker network create proxy
```

Verify that it exists:

```
docker network ls | grep proxy
```

The network must exist before the first docker compose up. Without it, both projects abort with the message "network proxy declared as external, but could not be found".

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4. Step 2 – Deploy the Nginx Proxy Manager

Create a dedicated directory for the reverse proxy on the Docker host and place the file docker-compose.yml inside it:

```
mkdir -p /opt/npm
cd /opt/npm
```

Contents of /opt/npm/docker-compose.yml:

```
services:
  npm:
    image: jc21/nginx-proxy-manager:2
    restart: unless-stopped
    ports:
      - "80:80"     # HTTP + ACME challenge
      - "443:443"   # HTTPS
      - "81:81"     # administration interface
    volumes:
      - npm_data:/data
      - npm_letsencrypt:/etc/letsencrypt
    networks:
      - proxy

networks:
  proxy:
    external: true
    name: proxy

volumes:
  npm_data:
  npm_letsencrypt:
```

Then start the container:

```
docker compose up -d
```

4.1 First Sign-In

The administration interface is then available at http://<docker-host>:81. On first start the Nginx Proxy Manager creates a default account whose credentials are documented with the image.

> Change the email address and password of the default account immediately after the first sign-in. As long as the default values apply, anyone with network access to port 81 can modify the certificates and forwarding rules of the entire environment.

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5. Step 3 – Deploy WolfISMS

The WolfISMS package contains docker-compose.yml as well as the template .envdefault. The actual configuration is done exclusively through environment variables in the file .env.

5.1 Obtain the Package and Create the Configuration

```
git clone https://github.com/wolfschus/wolfisms.git
cd WolfISMS
cp .envdefault .env
```

Then open .env and adapt the values to your environment. Chapter 8 contains the complete reference of all variables.

If you need your own certificates – for example to connect to an external database server – copy them into the directory certs/. This is not required for outward-facing TLS termination; the Nginx Proxy Manager takes care of that.

5.2 Start the Containers

```
docker compose up -d
```

Check the start-up:

```
docker compose ps
docker compose logs -f
```

5.3 Connecting to the Reverse Proxy

For the Nginx Proxy Manager to reach the application container, both must be on the proxy network and the container must be addressable under a stable name. That name is set through the variable PROXY_ALIAS and entered in the proxy host as the forward hostname. The supplied docker-compose.yml already joins the network; the following example shows the relevant sections:

```
services:
  wolfisms:
    # ... image, volumes, depends_on ...
    env_file: .env
    restart: unless-stopped
    networks:
      proxy:
        aliases:
          - ${PROXY_ALIAS}    # e.g. wolfisms-app

networks:
  proxy:
    external: true
    name: proxy
```

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No port mapping for the application:
The WolfISMS container needs no ports entry. It is addressed exclusively through the internal proxy network. An additionally published port would make the application reachable directly over HTTP, bypassing TLS termination – and, because of the Secure cookies, unusable for signing in as well.

The authoritative docker-compose.yml is part of the installation package. The example above shows only the sections relevant to the proxy connection and does not replace the supplied file.

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6. Step 4 – Configure the Proxy Host and Certificate

Publication is then configured in the NPM interface under Hosts → Proxy Hosts → Add Proxy Host.

6.1 "Details" Tab

Field	Value
Domain Names	isms.your-domain.com
Scheme	http
Forward Hostname	wolfisms-app (the name set in PROXY_ALIAS)
Forward Port	80
Block Common Exploits	enabled
Websockets Support	enabled

6.2 "SSL" Tab

- **SSL Certificate**: select "Request a new SSL Certificate"
- **Force SSL**: enable – redirects HTTP requests to HTTPS automatically
- **HTTP/2 Support**: enable
- **Email address**: provide it for the Let's Encrypt expiry notifications and accept the terms of service

For the standard challenge Let's Encrypt requires the domain name to resolve publicly to the host IP address and port 80 to be reachable from the outside. If that is not the case – for example in a purely internal environment – configure a DNS challenge with your DNS provider instead, or upload your own certificate.

[IMAGE PLACEHOLDER]

Figure: Creating a proxy host

Screenshot of the "Add Proxy Host" dialog in the Nginx Proxy Manager with the "Details" tab filled in

[IMAGE PLACEHOLDER]

Figure: Requesting a certificate

Screenshot of the "SSL" tab with the options "Request a new SSL Certificate", "Force SSL" and "HTTP/2 Support"

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6.3 Verify the Installation

Check	Expected result
Open https://isms.your-domain.com	The login page appears and the certificate is shown as valid
Request the same address over HTTP	Automatic redirect to HTTPS (Force SSL)
Sign in	Redirect to the dashboard – if signing in fails, see chapter 9
Container log	docker compose logs without traceback entries
Admin area	The "Updates" menu entry is absent – this is intended in container mode (see chapter 7)

Immediately after the first sign-in:
1) Change the default password of the admin user
2) Install the licence file customer.lic (Admin → Licence)
3) Restart the container so the licence and the secret key derived from it take effect: docker compose restart

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7. Operating in Container Mode

With the environment variable WOLFISMS_CONTAINER=1 the application recognises that it runs inside a container (config.container_mode). This changes its behaviour in two places.

7.1 In-App Updater Disabled

The entire update area including /admin/updates is disabled and responds with HTTP 404; the tile in the admin area is hidden. This is intentional: an update applied inside a running container would be lost the next time the image is restarted.

In Docker operation, updates are applied exclusively through a new image:

```
docker compose pull
docker compose up -d
```

7.2 Configuration Comes Only From the Environment

Outside Docker, config.py reads the credentials from the file /etc/wolfisms/secret.env. Inside the container that file usually does not exist – which is not an error: if it is missing, the application uses only the environment variables that are set. Those are exactly what .env provides through docker compose.

Different path:
If you do want to use an env file inside the container, its path can be set with the variable WOLFISMS_ENV_FILE. Environment variables that are already set always take precedence over the values from the file.

7.3 Data Persistence and Backup

The database and the upload directory must be persisted. Both have to be mounted as volumes, otherwise they are lost when the container is replaced.

Data	Location	Note
Database	Volume of the database container or external database server	Contains all ISMS content
Uploaded files	UPLOAD_DIR (default /var/www/isms/wolfisms/uploads)	Attachments, playbooks, logo, templates and letterhead
Sessions	SESSION_DIR (default /var/www/isms/wolfisms/sessions)	Must be writable; losing it merely signs all users out. The application sets the directory to 0700 and the session files to 0600 at startup
Licence file	license/customer.lic in the application directory	After switching images, check that it is still present

The backup itself works as in the classic installation, through the admin area at /admin/backup or the direct link /backup/download. The archive contains the database dump and the upload

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directory; application code, configuration and the licence file are not included and must be backed up separately. Chapter 5 of the administration documentation covers the details.

7.4 Logs

Purpose	Command
Follow live output	docker compose logs -f
Application only	docker compose logs -f wolfisms
Last 200 lines	docker compose logs --tail=200
Proxy access and certificates	docker compose logs -f npm (in the directory /opt/npm)

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8. Configuration Reference (.env)

All parameters are set as environment variables. They are resolved in this order: environment variables that are already set, then the env file, and finally the defaults defined in the code. There is deliberately no default for DB_PASSWORD – if the variable is missing, the application will not start.

8.1 Mandatory Settings

Variable	Example	Meaning
DB_PASSWORD	(strong password)	Database password – the only value without a default
DB_USER	isms	Database user (default: wolf)
DB_HOST	db	Host name of the database server; the service name within the Compose project
DB_NAME	isms	Name of the database
ISMS_BASE_URL	https://isms.your-domain.com	Public base URL – required for links in reminder emails
FLASK_SECRET_KEY	(64 hex characters)	Fallback session key. If a valid licence is present it is ignored and the key is derived from the licence ID
WOLFISMS_CONTAINER	1	Enables container mode (see chapter 7)
PROXY_ALIAS	wolfisms-app	Network alias of the container; enter it as the forward hostname in the proxy host

Generate a secure random value for FLASK_SECRET_KEY:

```
python3 -c "import secrets; print(secrets.token_hex(32))"
```

8.2 Paths and Instance

Variable	Default	Meaning
WEBPATH	(empty)	URL path prefix. Leave empty when running under a dedicated domain; set it only if the application runs under a sub-path such as /isms
PATHNAME	isms	Directory name of the instance
UPLOAD_DIR	/var/www/isms/wolfisms/uploads	Path for uploaded files – mount as a volume
SESSION_DIR	/var/www/isms/wolfisms/sessions	Path for server-side sessions – must be writable
WOLFISMS_ENV_FILE	/etc/wolfisms/secret.env	Alternative path of an env file inside the container

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8.3 Language

Variable	Default	Meaning
DEUTSCH	True	Allow the German user interface
ENGLISCH	False	Allow the English user interface. With DEUTSCH=0 and ENGLISCH=1 the installation runs in English only

8.4 Single Sign-On (optional)

Variable	Meaning
MS_AUTH	1 = enable sign-in through Microsoft Entra ID
GOOGLE_AUTH	1 = enable sign-in through Google
SSO_ALLOWED_DOMAINS	Comma-separated allow list of permitted email domains. An empty list admits nobody – that is the safe default
SSO_AUTO_CREATE	1 = create unknown SSO users automatically (default 0)
MS_TENANT_ID / MS_CLIENT_ID / MS_CLIENT_SECRET / MS_REDIRECT_URI	Credentials and callback URL for Microsoft
GOOGLE_CLIENT_ID / GOOGLE_CLIENT_SECRET / GOOGLE_REDIRECT_URI	Credentials and callback URL for Google

The callback URL must point to the public host name served by the reverse proxy – not to the container name.

8.5 Further Variables

Variable	Meaning
UPDATE_PROXY	Proxy for outbound connections. In container mode it has no effect on the disabled updater
RATELIMIT_STORAGE_URI	Storage for the rate limit counters. The default memory:// counts per worker process – with several workers every limit is multiplied accordingly. For such installations point it at shared storage, e.g. redis://redis:6379
ISMS_DEBUG	1 = switch on Flask debug mode. For troubleshooting only; remove it afterwards

ISMS_DEBUG must not remain set in production. Debug mode enables PROPAGATE_EXCEPTIONS, disables template caching and is a finding in every hardening baseline.

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9. Troubleshooting

Symptom	Cause and remedy
"network proxy declared as external, but could not be found"	The Docker network is missing. Create it with docker network create proxy (chapter 3) and start the containers again
NPM reports "502 Bad Gateway"	The application container cannot be reached under the configured forward hostname. Check that the container is running, that it is on the proxy network, and that PROXY_ALIAS matches the name entered in the proxy host
The login page appears but signing in fails silently	You are accessing the site over HTTP instead of HTTPS. Session cookies are Secure=True and are not set over unencrypted connections. Enable "Force SSL" in the proxy host
The container does not start and the log mentions the database	DB_PASSWORD is missing or the database is not ready yet. The variable is mandatory and deliberately has no default; check the start order via depends_on
The certificate request fails	The DNS record does not point to the host, or port 80 is not reachable from the outside. Check both or switch to a DNS challenge
Links inside the ISMS point to http:// or the container name	The proxy does not pass X-Forwarded-Proto or X-Forwarded-Host, or ISMS_BASE_URL is set incorrectly. Check both
The "Updates" menu entry is missing in the admin area	Not an error: in container mode the in-app updater is disabled. Update through a new image (chapter 7.1)
Demo mode persists after installing the licence	The licence check is cached. Restart the container: docker compose restart
Uploaded files disappeared after an update	UPLOAD_DIR was not mounted as a volume. Add the volume and restore the files from the most recent backup

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10. Checklist

✓	Task
☐	DNS record resolves to the public IP address of the Docker host
☐	Ports 80 and 443 are reachable from the outside
☐	Port 81 is restricted to the internal network
☐	Docker network proxy created
☐	Nginx Proxy Manager started, default credentials changed
☐	.env created from .envdefault and fully populated
☐	DB_PASSWORD set and FLASK_SECRET_KEY replaced with a random value
☐	WOLFISMS_CONTAINER=1 set
☐	PROXY_ALIAS set and entered as the forward hostname in the proxy host
☐	ISMS_BASE_URL set to the public host name
☐	Volumes mounted for the database and UPLOAD_DIR
☐	Proxy host created, certificate requested, Force SSL and HTTP/2 enabled
☐	Sign-in over HTTPS tested successfully
☐	Default password of the admin user changed
☐	Licence file installed and the container restarted
☐	Second admin account created as emergency access
☐	Backup via /admin/backup tested and stored off-site
☐	ISMS_DEBUG not set
☐	.env not committed to a Git repository

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11. Further Documentation

Document	Content
WolfISMS Administration Documentation	Complete system administration: user and permission management, backup, licence management, REST API, two-factor authentication, single sign-on, AI settings and security notes. It also explains what differs in container mode compared with the classic installation on Apache with mod_wsgi
WolfISMS User Documentation	Description of all functional modules for end users
WolfISMS REST API	Interface description for connecting external systems: incidents, measures, assets, risks and suppliers – with all fields, filters, value lists and a complete example sync client

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